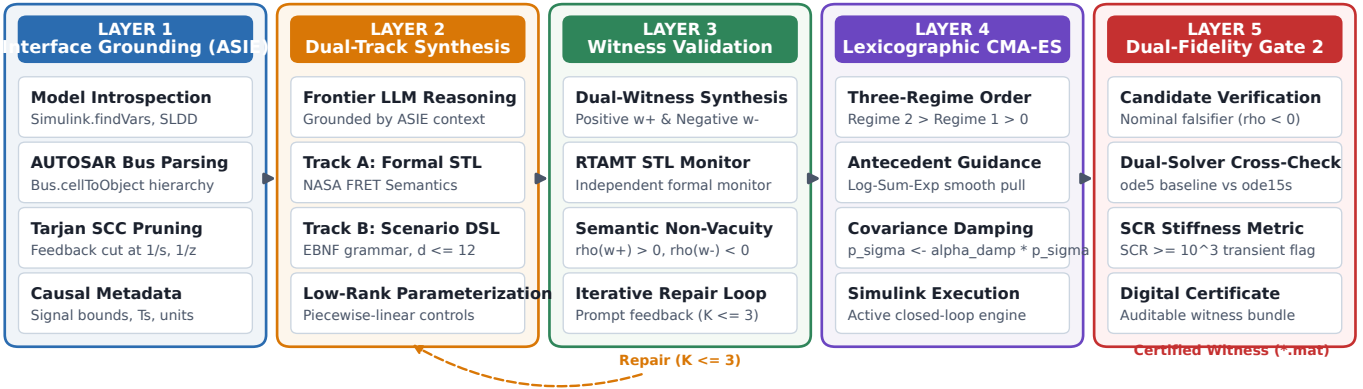
LAYER 1
terface Grounding (ASIE
Model Introspection
Simulink.findVars, SLDD
AUTOSAR Bus Parsing
Bus.cellToObject hierarchy
Tarjan SCC Pruning
Feedback cut at 1/s, 1/z
Causal Metadata
Signal bounds, Ts, units
LAYER 2
Dual-Track Synthesis
Frontier LLM Reasoning
Grounded by ASIE context
Track A: Formal STL
NASA FRET Semantics
Track B: Scenario DSL
EBNF grammar, d <= 12
Low-Rank Parameterization
Piecewise-linear controls
LAYER 3
Witness Validation
Dual-Witness Synthesis
Positive w+ & Negative w-
RTAMT STL Monitor
Independent formal monitor
Semantic Non-Vacuity
rho(w+) > 0, rho(w-) < 0
Iterative Repair Loop
Prompt feedback (K <= 3)
LAYER 4
Lexicographic CMA-ES
Three-Regime Order
Regime 2 > Regime 1 > 0
Antecedent Guidance
Log-Sum-Exp smooth pull
Covariance Damping
p_sigma <- alpha_damp * p_sigma
Simulink Execution
Active closed-loop engine
LAYER 5
Dual-Fidelity Gate 2
Candidate Verification
Nominal falsifier (rho < 0)
Dual-Solver Cross-Check
ode5 baseline vs ode15s
SCR Stiffness Metric
SCR >= 10^3 transient flag
Digital Certificate
Auditable witness bundle
Repair (K <= 3)
Certified Witness (*.mat)